

Nathan Labiosa

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Summary

ML engineer who ships models into production, built a multi-agent ML pipeline at Visa that lifted production model performance 2–10% across team runs, deployed eight drug-activity models into a live product, and distilled a 65B multimodal model to 7B at 92% lower compute. NeurIPS 2025 author.

Skills

ML Frameworks: PyTorch, DeepSpeed (distributed training), Hugging Face, TensorFlow, Scikit-learn, LangChain, LangGraph

Infrastructure & Serving: AWS, GCP, Docker, Linux, Git, ETL pipelines, MLOps

Data & Retrieval: SQL (PostgreSQL, SQLite), vector databases, Neo4j/Cypher, RAG, GraphRAG

Languages: Python, C++, SQL, Bash

Domains: Computer Vision, NLP, RL, Embeddings and Retrieval, Multimodal Models

Education

University of Southern California

Master of Science - Computer Science

Los Angeles, CA

Sep 2025 - May 2027

University of Wisconsin - Madison

Bachelor of Science - Biomedical Engineering and Computer Science

Madison, WI

May 2025

Experience

Visa

Austin, TX

Machine Learning Engineer Intern

May 2026 - Aug 2026

- Built and deployed an autonomous, containerized DS pipeline (10-agent LangGraph DAG via FastAPI and ChromaDB) to Visa's Sentinel Grid Kubernetes platform, compressing 1 week of data-scientist workflow into 30 minutes with human-in-the-loop review gates.
- Designed a dual-write experiment knowledge base (SQLite + ChromaDB, 11 vector collections) that surfaces cross-team modeling priors, improving downstream model performance 2–10% and cutting redundant pipeline iterations 2x.
- Implemented zero-overhead feature attribution via counterfactual metric deltas computed in SQL against historical runs, measuring feature effectiveness without in-run ablations.
- Diagnosed three production blockers (single-writer DB serialization, synchronous training blocking the async event loop, per-process event bus) and leading migration to a PostgreSQL pool, dedicated REST compute service, and pub/sub events.

Advanced Space

Westminster, CO

Machine Learning Intern

May 2025 - Aug 2025

- Built an automated ETL pipeline using Python and Neo4j to ingest and structure 5,000+ technical documents, reducing data retrieval latency by 30%
- Developed a GraphRAG system by building a modular MCP tool that integrates Cypher query generation; outperformed baseline vector search accuracy by 18% on complex, multi-hop queries.
- Created and optimized RL/IRL algorithms for multi-object simulation environments, achieving >85% success rates in latent goal inference through hyperparameter tuning and simulation testing.
- Contributed technical architecture and feasibility analysis for a NASA NIAC Phase 1 proposal, defining ML system requirements for autonomous space exploration.

Revilico

Los Angeles, CA

Machine Learning Engineer

Apr 2024 - May 2025

- Architected and managed model training pipelines on AWS, optimizing compute resources to validate eight novel drug activity predictors.
- Standardized model inference scripts and artifacts, streamlining the handoff to DevOps and enabling seamless integration of models into the user-facing frontend.
- Engineered feature extraction tools for SMILES sequences, boosting IC50/EC50 prediction R^2 scores by 0.12 through improved data representation.
- Designed deep learning architectures for proteomic classification, improving accuracy by 15-25% over industry baselines across 30 protein families.
- Built RL frameworks for de novo molecular generation, yielding 100+ novel compounds optimized for electronegativity, folding patterns, and pH balance.

Research Experience

University of Southern California

Los Angeles, CA

Research Assistant in Physical Superintelligence Lab

August 2025 - Present

- Training a $\pi 0.5$ vision-language-action model on ~30M frames from DROID and other manipulation datasets for in-context adaptation to unseen robot embodiments.
- Developed an embodiment-conditioning method that predicts action-normalization parameters from ~20 frames of motion context (same- and cross-episode), enabling a single policy to adapt its action distribution to new kinematics without retraining.

University of Wisconsin - Madison

Madison, WI

Computer Vision Researcher

May 2024 - May 2025

- Contributed to a novel framework for temporally stable and corruption-robust video inference, integrating stabilization adapters into frozen vision backbones.
- Headed design and evaluation of adversarial robustness experiments, delivering +11.8% accuracy under iterative attacks without retraining base networks.
- Crafted stabilizer modules for ResNet architectures, reducing fine-tuning time by 40% while maintaining state-of-the-art robustness.
- Constructed and benchmarked a binary human/nonhuman classification task on the DAVIS dataset (3,455 annotated frames) with a fine-tuned ResNet-50, optimizing training schedules and augmentation protocols.

University of Central Florida

Orlando, FL

Visual Large Language Model Researcher

May 2023 - Oct 2023

- Oversaw collaboration with researchers from the University of Central Florida and Meta, driving significant advancements in large language model research.
- Refined complex, 65 billion parameter AI models into streamlined, 7 billion parameter university-level models, reducing compute cost by 92% and boosting widespread research applicability.
- Devised a visual large language model (VLLM) training method using novel techniques, merging over 150 thousand images and 160 thousand text examples during multiple training stages.
- Attained 95% of industry-leading text-based models leveraging a practical sized VLLM.
- Utilized Pytorch and DeepSpeed frameworks to execute parallel training in a high-performance GPU cluster

University of Wisconsin - Madison

Madison, WI

Quantitative Cell Imaging Researcher

Feb 2023 - Sep 2023

- Constructed novel software tools for tumor micro-environment quantification employing deep learning in the LOCI research laboratory, attaining 95% IOU vs state-of-the-art models.
- Improved cell recognition throughput by 1.2x, enabling lab imaging pipelines to analyze ~10,000 cells/day.

Publications

Sensitivity, Causality, and Repair Dissociate: A Layer-Wise Analysis of Perturbation Robustness and Its Scaling

Labiosa, N., Buff, D. Nayak, E., and Donno, E.

ARR Under Review. doi: 10.48550/arXiv.2608.03842

Instant Video Models: Universal Adapters for Stabilizing Image-Based Networks

Dutson M., **Labiosa, N.**, Li, Y., and Gupta, M.

NeurIPS, 2025. doi: 10.48550/arXiv.2512.03014

Hybrid Transformer Model for Protein Classification

Labiosa, N., A. Kohli, C. Chung, and C. Korban.

bioRxiv, 2024. doi: 10.1101/2024.10.31.621421

Visual Information and Large Language Models: A Deeper Analysis

Labiosa, N., Huynh, D., Lim, S.

2023 UCF CRCV Catalog